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A Monte Carlo investigation of cumulative dose measurements for cone beam computed tomography (CBCT) dosimetry

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Abstract

Many studies have shown that the computed tomography dose index ($CTDI_{100}$) which is considered as a main dose descriptor for CT dosimetry fails to provide a realistic reflection of the dose involved in cone beam computed tomography (CBCT) scans. Several practical approaches have been proposed to overcome drawbacks of the $CTDI_{100}$. One of these is the cumulative dose concept. The purpose of this study was to investigate four different approaches based on the cumulative dose concept: the cumulative dose (1) $f(0,150)$ and (2) $f(0,\infty)$ with a small ionization chamber 20 mm long, and the cumulative dose (3) $f_{100}(150)$ and (4) $f_{100}(\infty)$ with a standard 100 mm pencil ionization chamber. The study also aimed to investigate the influence of using the 20 and 100 mm chambers and the standard and the infinitely long phantoms on cumulative dose measurements. Monte Carlo EGSnrc/BEAMnrc and EGSnrc/DOSXYZnrc codes were used to simulate a kV imaging system integrated with a TrueBeam linear accelerator and to calculate doses within cylindrical head and body PMMA phantoms with diameters of 16 cm and 32 cm, respectively, and lengths of 150, 600, 900 mm. $f(0,150)$ and $f_{100}(150)$ approaches were studied within the standard PMMA phantoms (150 mm), while the other approaches $f(0,\infty)$ and $f_{100}(\infty)$ were within infinitely long head (600 mm) and body (900 mm) phantoms. $CTDI_\infty$ values were used as a standard to compare the dose values for the approaches studied at the centre and periphery of the phantoms and for the weighted values. Four scanning protocols and beams of width 20–300 mm were used. It has been shown that the $f(0,\infty)$ approach gave the highest dose values which were comparable to $CTDI_\infty$ values for wide beams. The differences between the weighted dose values obtained with the 20 and 100 mm chambers were significant for the beam widths <120 mm, but these differences declined with increasing beam widths to be within 4%. The weighted dose values calculated within the infinitely long phantoms with both the chambers for the beam widths ≤140 were within 3% of those within the standard phantoms, but the differences rose to be within 15% at wider beams. By comparing the approaches studied in this investigation with other methodologies taking into account the efficiency of the approach as a dose descriptor and the simplicity of the implementation in the clinical environment, the $f(0,150)$ method may be the best for CBCT dosimetry combined with the use of correction factors.

Keywords: cumulative dose, CBCT, CTDI, EGSnrc/BEAMnrc, EGSnrc/DOSXYZnrc, AAPM TG-111

(Some figures may appear in colour only in the online journal)

1. Introduction

Cone beam computed tomography (CBCT) is based on the use of a wide x-ray beam with a flat panel detector. Since its introduction, CBCT has developed a major role in medical applications such as image guided radiation therapy (IGRT) that helps to deliver a prescribed dose to a cancer patient with a high degree of precision, as well as offering an additional imaging option in interventional radiology. However, a dosimetry system to monitor performance of CBCT scanners in terms of the output dose is required, as there are drawbacks in that used for conventional CT scanners that employ narrow beams. For many years, the CT dose index (CTDI) concept has been utilized as the main dose descriptor for CT scanner performance (Shope *et al* 1981). The $CTDI_{100}$ characterizes a CT scanner by integrating a dose profile resulting from a single axial rotation using a standard pencil ionization chamber with an active length of 100 mm set up at the isocentre parallel to the rotation axis (IEC 2001). The $CTDI_{100}$ is also used as an indicator for the dose received by a patient undergoing a CT scan by performing the dose measurements within standard cylindrical polymethyl methacrylate (PMMA) phantoms of length 150 mm with diameters of 16 cm representing an adult head or a paediatric body and 32 cm representing an adult body. The dose measurements within these phantoms are made on the central axis and at peripheral positions 1 cm below the phantom surfaces. However, as the main concept of CBCT scans is to use an x-ray beam which is usually wider than the length of the 100 mm ionization chamber and sometimes wider than the standard 150 mm long PMMA phantoms, the $CTDI_{100}$ concept is no longer appropriate for CBCT scans. Studies of the suitability of utilizing the $CTDI_{100}$ concept for CBCT have shown that the $CTDI_{100}$ fails to provide a realistic reflection of the dose (Mori *et al* 2005, Boone 2007, Kyriakou *et al* 2008, Geleijns *et al* 2009, Abuhaimed *et al* 2014). The shortcoming is its inability to accommodate and record the whole primary beam and the scattered radiation. To overcome the $CTDI_{100}$ limitation, various dosimetry approaches have been investigated (Mori *et al* 2005, Islam *et al* 2006, Fahrig *et al* 2006, Amer *et al* 2007, Kyriakou *et al* 2008, Geleijns *et al* 2009, AAPM 2010, Dixon and Boone 2010, IEC 2010).

Dixon (2003) has suggested a dosimetry approach to replace the $CTDI_{100}$ concept for conventional CT scans based on measuring the cumulative dose for a CT scan using a small ionization chamber positioned at the centre ($z = 0$) of a phantom that is long enough to provide a full scatter condition, i.e. further extending the length of the phantom will have a negligible effect on the measurement. The measurement of the cumulative dose is comparable to the multiple scan average dose (MSAD) which can be estimated from the $CTDI_{100}$ for conventional CT scans. The American Association of Physicists in Medicine (AAPM) TG–111 has adapted the cumulative dose concept and proposed it for dosimetry of conventional CT and CBCT scans (AAPM 2010). This requires the use of a small ionization chamber within a long phantom made of PMMA, water, or polyethylene. Fahrig *et al* (2006) have assumed that the contribution of scattered radiation arising from longer phantoms at the central point at which the cumulative dose is measured is minimal, and so have proposed that the cumulative dose for CBCT scans can be measured within the standard PMMA phantoms with minimal differences from results with longer phantoms. This approach has been utilized to evaluate dose for CBCT scans in various studies (Song *et al* 2008, Osei *et al* 2009, Kim *et al* 2010b, Cheng *et al* 2011).

Amer *et al* (2007) have suggested replacing the $CTDI_{100}$ concept for CBCT scans by an approach called a cone beam dose index (CBDI). This requires measurement of the cumulative dose using a standard 100 mm pencil ionization chamber within the standard PMMA phantoms. Geleijns *et al* (2009) have investigated this approach for a beam of width 160 mm and found measurements using the standard 100 mm ionization chamber within the standard PMMA phantoms to be within 10% of the $CTDI_{300}$ measured with a 300 mm pencil chamber within head and body phantoms 350 mm in length.

The International Electrotechnical Commission (IEC) has also proposed an approach for CBCT dosimetry involving an extension of the $CTDI_{100}$ concept (IEC 2010). This approach utilizes the $CTDI_{100}$ dosimetry equipment available in the majority of hospitals. It is based on the acquisition of $CTDI_{100}$ measurements for a reference beam of width ≤40 mm within the standard PMMA phantoms to which a correction factor is applied, equal to the ratio of $CTDI_{100}$ measurements free in air for the wide beam of interest and the reference beam (IEC 2010, IAEA 2011, Platten *et al* 2013). The efficiency of this approach in accounting for all the scattered radiation occurring during CBCT scans has been shown through Monte Carlo (MC) simulations to be independent of beam width (Abuhaimed *et al* 2014).

The MC simulation technique is a valuable tool for assessing not only the doses received by patients undergoing CT or CBCT scans, but also dose distributions within phantoms for evaluation of the performance of different dosimetry techniques. EGSnrc code (Kawrakow 2000, Kawrakow and Rogers 2000) is one of the more common MC codes being utilized to simulate medical applications for MV and kV energies. Although EGSnrc code is well known to simulate linear accelerators in the range MV energies, it has been benchmarked to simulate kV applications (Mainegra-Hing and Kawrakow 2006, Ali and Rogers 2007, 2008). The EGSnrc-based user codes known as BEAMnrc (Rogers *et al* 1995) and DOSXYZnrc (Walters *et al* 2013) have played a critical role in evaluation of dosimetry for CBCT scans (Jarry *et al* 2006, Ding *et al* 2007, Chow *et al* 2008, Downes *et al* 2009, Ding and Coffey 2009, Kim *et al* 2010a, Qiu *et al* 2012, Ding and Munro 2013).

The aim of the present study was to investigate four approaches based on the cumulative dose concept for CBCT dosimetry using different scanning protocols and a range of beam widths. MC simulations were used to study the four approaches: the cumulative dose (1) $f(0,150)$ and (2) $f(0,\infty)$ with a small ionization chamber 20 mm long, and the cumulative dose (3) $f_{100}(150)$ and (4) $f_{100}(\infty)$ with a standard 100 mm pencil ionization chamber. The $f(0,150)$ and $f_{100}(150)$ were studied within the standard PMMA phantoms, while $f(0,\infty)$

and $f_{100}(\infty)$ were within infinitely long head and body phantoms. In addition, the influence of using the 20 mm and 100 chambers on the cumulative dose measurements was investigated. Contributions resulting from the primary beam and scattered radiation created under a scatter equilibrium condition to the cumulative dose measurements were investigated. The imaging doses involved in CBCT scans for head and body scanning protocols were evaluated experimentally using the three approaches $f(0,150)$, $f_{100}(150)$, and $f(0,\infty)$, and the results were compared with other methodologies. The study also aimed to compare the various dosimetric approaches, consider whether they were appropriate for reporting the doses received by patients, and assess the simplicity of implementation in the clinical environment.

2. Materials and methods

2.1. kV system and scanning protocols

A kV imaging system integrated with a Varian TrueBeam linear accelerator (Varian Medical systems, Palo Alto, CA) was employed both for practical measurements and the simulations in this study. The kV imaging system can be operated with tube voltages between 40 and 140 kV, and has four collimator blades X1, X2, Y1, and Y2 that work independently. This enables delivery of a symmetrical or an asymmetrical field with minimum and maximum field sizes of 2 cm × 2 cm to 50 cm × 50 cm. The collimator blades X1 and X2 set the scan diameter, while Y1 and Y2 select the extent of the scan in the axial direction. Unlike conventional CT scans, CBCT scans are acquired only using the axial scan mode with a fixed patient table position. In addition to a number of internal filters, two different types of bowtie filter, full and half, are used to improve the quality of the CBCT images by reducing the range of x-ray intensities to be recorded.

The kV system can be employed with either Full-Fan or Half-Fan acquisition modes, where the choice of the mode is based on the size of the region of interest. The field of views (FOVs) for the full-fan and half-fan modes are 26.4 cm and 47.8 cm at the isocentre respectively. The full-fan mode is acquired through either a full 360° scan or a half 200° scan with the full bowtie filter. A symmetrical field size is used for the Full-Fan mode, with collimator blades X1 and X2 being set up at 13.2 cm, and Y1 and Y2 at 9.9 cm at the isocentre. The flat panel detector is placed at 150 cm from the kV source and is set so that its centre matches that of the region to be imaged. The Half-Fan mode provides a larger FOV by shifting the flat panel detector laterally by 16 cm, and using an asymmetrical field. The collimator blades X1 and X2 are opened by 2.6 cm and 23.9 cm at the isocentre respectively, while the Y1 and Y2 blades are set similar to those used for the full-fan mode. Only the full 360° scan is used for the half-fan mode. More details for the kV imaging system involved in the present investigation were given in a previous study (Abuhaimed *et al* 2014).

In the clinic, five CBCT scanning protocols are used head, thorax, thorax slow, pelvis, and pelvis spot light. Each scanning protocol is employed with parameters (kV, mAs, number of projections, etc.) that are based on those provided by the manufacturer. In order to ensure that all scanning parameters used in the clinic were covered, four scanning protocols were defined and investigated in this study namely Head-HS, Head-FS, Body-HS, and Body-FS (table 1). Head-HS and Head-FS were defined to represent the head scan with the half 200° scan (HS) and the full 360° scan (FS) modes, while Body-HS covered the pelvic spot light scan with the half 200° scan (HS) mode, and Body-FS was used to investigate the pelvic and thorax scans with the full 360° scan (FS) mode.

Table 1. Scanning protocols employed in this study.

	Body-HS	Body-FS	Head-HS	Head-FS
Tube voltage (kV)	125	125	100	100
X-ray current (mA)	80	20	20	20
X-ray millisecond (ms)	25	20	20	20
Exposures (mAs)	733	264	147	264
Acquisition mode	Full-Fan	Half-Fan	Full-Fan	Full-Fan
Gantry rotation (degrees)	200°	360°	200°	360°
Left side-90°, right side-270°	90–290°	0–360°	90–290°	0–360°
The nominal beam width (mm)		20–300		
X1 and X2 (mm)	132, 132	25, 239	132, 132	132, 132
Y1 and Y2 (mm)		(10–150), (10–150)		

2.2. CBCT dosimetry

The $CTDI_{100}$ is defined as:

$$CTDI_{100} = \frac{1}{N \times T} \int_{-50\text{ mm}}^{50\text{ mm}} D(z)\ dz, \tag{1}$$

where N is the number of slices obtained in a single axial scan, T is the nominal width of a single slice, and $D(z)$ is the dose profile along the axis of a single axial rotation. The $CTDI_{100}$ measurements are performed at the centre and at the four peripheral positions within the standard PMMA phantoms set up at source-isocentre-distance (SID) = 100 cm. In CBCT scans, the region of interest is scanned with a single detector and a single rotation. Therefore, the nominal beam width of the scan (W) is used instead of ($N \times T$) in equation (1). In order to take the variation in dose within the phantom into account, a weighted $CTDI_{100}$ ($CTDI_w$) is defined as:

$$CTDI_w = \frac{1}{3} CTDI_{100,c} + \frac{2}{3} CTDI_{100,p}, \tag{2}$$

where $CTDI_{100,c}$ is measured at the centre of the phantom and $CTDI_{100,p}$ is calculated as an average of four measurements taken at the peripheral positions. In order to quantify the total amount of radiation to which a patient is exposed during a CT or CBCT scan, the $CTDI_\infty$ concept can be used. $CTDI_\infty$ takes into account all the contributions resulting from the primary beam and the scattered radiation by quantifying the dose from a scan within an infinitely long phantom as:

$$CTDI_\infty = \frac{1}{W} \int_{-\infty}^{\infty} D(z)\, dz\,. \tag{3}$$

The cumulative dose for a CBCT scan is measured at the middle of the standard PMMA phantoms $f(0,150)$ or at the middle of an infinitely long phantom $f(0,\infty)$ set up at SID = 100 cm as a point dose for the peak value of the dose profile using a small ionization chamber such as a standard 0.6 cm^3 Farmer chamber with an active length of 20 mm (Fahrig *et al* 2006, AAPM 2010). In order to estimate $f(0,150)$ or $f(0,\infty)$ for a CBCT scan of multiple rotations (N), the point dose measured for a single scan is multiplied by the number of the rotations.

However, at the present time only a single CBCT scan is acquired to obtain a scan for the patient ($N = 1$).

Similar to CTDI_{w}, weighted $f(0,150)_{\text{w}}$ and $f(0,\infty)_{\text{w}}$ can be defined as:

$$f(0,150)_{\text{w}} = \frac{1}{3} f(0,150)_{\text{c}} + \frac{2}{3} f(0,150)_{\text{p}}\,, \tag{4}$$

and

$$f(0,\infty)_{\text{w}} = \frac{1}{3} f(0,\infty)_{\text{c}} + \frac{2}{3} f(0,\infty)_{\text{p}}\,. \tag{5}$$

The cumulative dose $f_{100}(150)$ and $f_{100}(\infty)$ are measured as an average dose over the 100 mm length of the standard pencil ionization chamber within the standard PMMA phantoms and infinitely long phantoms respectively. $f_{100}(150)$ and $f_{100}(\infty)$ are defined as (Amer *et al* 2007, Geleijns *et al* 2009):

$$f_{100}(150) = f_{100}(\infty) = \frac{1}{100} \int_{-50\ \text{mm}}^{50\ \text{mm}} D(z)\ \text{d}z, \tag{6}$$

and the weighted $f_{100}(150)_{\text{w}}$ and $f_{100}(\infty)_{\text{w}}$ are evaluated in a similar manner to that in equations (4) and (5).

Mori *et al* (2005) and Kyriakou *et al* (2008) have shown that the CTDI concept can be extended for CBCT dosimetry by using a long ionization chamber within longer PMMA phantoms to estimate CTDI_{∞} which provides a measurement related to the total dose to the whole body involved in a CT or CBCT scan. Dixon and Boone (2010) have shown that the cumulative dose resulting from a single axial CBCT scan for a beam of width W is comparable to that resulting from five axial scans each of width $\approx W/5$. Therefore, AAPM (2010) and Dixon and Boone (2010) have suggested the use of the cumulative dose concept, as it is an analogy to the CTDI concept and more practical than other proposed approaches. To investigate the relative suitability of the various proposed dose quantities for CBCT dosimetry, values for $f(0,150)$, $f(0,\infty)$, $f_{100}(150)$ and $f_{100}(\infty)$ in phantoms of varying size have all been compared with the respective CTDI_{∞} values in order to give an indication of their capability for reporting the 'total dose' used in a scan. The dose quantities were compared for the centre and periphery of the phantoms, and for the weighted values e.g. $f(0,150)_{\text{c}}/\text{CTDI}_{\infty,\text{c}}$, $f(0,150)_{\text{p}}/\text{CTDI}_{\infty,\text{p}}$, $f(0,150)_{\text{w}}/\text{CTDI}_{\infty,\text{w}}$.

In this study, 600 mm and 900 mm were used to represent infinite lengths for the head and the body phantoms respectively, (Mori *et al* 2005, Kyriakou *et al* 2008, Kim *et al* 2011, Li *et al* 2014):

$$\text{CTDI}_{\infty}(\text{head}) = \frac{1}{W} \int_{-300}^{300} D(z)\,\text{d}z \quad \text{and} \quad \text{CTDI}_{\infty}(\text{body}) = \frac{1}{W} \int_{-450}^{450} D(z)\,\text{d}z, \tag{7}$$

where W ranged from 20 to 300 mm with an increment of 20 mm. Figure 1 and table 2 summarize the methods used to calculate the four quantities $f(0,150)$, $f(0,\infty)$, $f_{100}(150)$ and $f_{100}(\infty)$ using the scanning protocols listed in table 1.

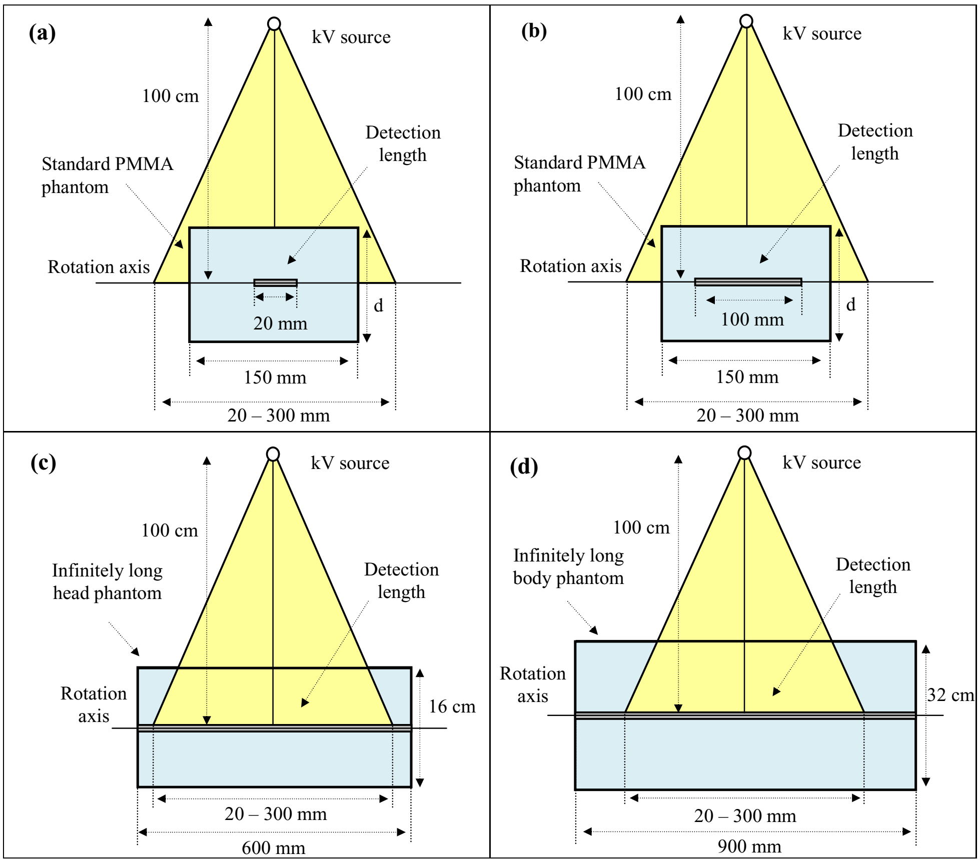

Figure 1. Diagrammatic representations of the configurations used to assess (*a*) $f(0,150)$, (*b*) $f_{100}(150)$, for both head and body phantoms and (*c*) CTDI_∞ within the infinitely long head phantom, and (*d*) CTDI_∞ within the infinitely long body phantom. The phantoms shown in (*c*) and (*d*) were also used for $f(0,\infty)$ and $f_{100}(\infty)$, but with detection lengths similar to those in (*a*) and (*b*).

Table 2. The PMMA phantoms used in this study.

Phantom	Diameter (*d*) (cm)	Length (mm)	Used for
Standard head	16	150	$f(0,150)$, $f_{100}(150)$
Standard body	32	150	$f(0,150)$, $f_{100}(150)$
Infinitely long head	16	600	$f(0,\infty)$, $f_{100}(\infty)$, CTDI_∞
Infinitely long body	32	900	$f(0,\infty)$, $f_{100}(\infty)$, CTDI_∞

2.3. MC simulations

The dose calculations for the quantities investigated were obtained using the user codes BEAMnrc and DOSXYZnrc. In the first step, the kV imaging system described in section 2.1 was modelled in detail using BEAMnrc. The materials and the geometrical specifications for the components of the system were obtained from the manufacturer. The simulation method for the kV imaging system has been described in detail in (Abuhaimed *et al* 2014). In order to achieve as low statistical uncertainty as possible, 1×10^9 histories were run for each simulation. ISOURCE = 10 of BEAMnrc was employed to generate mono-energetic electron beams of diameter of 1 mm and energies of 100 and 125 keV to hit the anode from the side with an angle of 14°. The low energy thresholds for creation of secondary electrons (AE) and the cut-off energies for transport of the electrons (ECUT) were set to 0.516 MeV, and for photons (AP) = (PCUT) was set to 0.001 MeV (Ding and Munro 2013). The collimator blades in the x and y axes for each scanning protocol were set to the parameters shown in table 1. A variance reduction technique (VRT) known as the directional bremsstrahlung splitting was used with a splitting number (NBRSP = 2×10^4) to improve the efficiency of the simulations (Kawrakow *et al* 2004, Mainegra-Hing and Kawrakow 2006). The default value of the boundary tolerance parameter $BDY_TOL was modified from (1×10^{-5}) to (5×10^{-7}) to enhance the MC calculations (Ali and Rogers 2008). The boundary crossing algorithm (EXACT) and electron step algorithm (PRESTA-II) were employed. The data of the National Institute of Standards and Technology (NIST), Koch–Motz, and NIST xcom were used for bremsstrahlung cross-sections, bremsstrahlung angular sampling, and photon cross-sections, respectively (Mainegra-Hing and Kawrakow 2006). Spin effect, the electron impact ionization, Rayleigh scattering, and atomic relaxations were turned on (Ding and Coffey 2010). The output for each BEAMnrc simulation was recorded in a phase space (PS) file at a source-surface-distance (SSD) = 75 cm.

In the second step: the resulting PS files were transferred and used as sources in DOSXYZnrc using ISOURCE = 8. The gantry rotation for each scanning protocol was set as indicted in table 1, and the distance between the source (the PS files) and the isocentre was set to 25 cm. The four cylindrical phantoms made of PMMA (ρ = 1.19 g cm^{-3}) given in table 2 were simulated with DOSXYZnrc to calculate the absorbed dose. The phantoms were simulated using voxels of different sizes ($0.5 \times 0.5 \times 0.5$ cm^3) at the centre and periphery of the phantoms and larger voxels at the middle, as Babcock *et al* (2008) have shown that using this technique results in minimizing time of simulations carried out with a homogenous phantom without affecting the dose accuracy. ($24–30 \times 10^8$) histories with a photon splitting number set to 300 were run to obtain a statistical uncertainty of less than 3%. The cross sectional data and the (ECUT) and (PCUT) values used in BEAMnrc were also involved in DOSXYZnrc. The HOWFARLESS transport algorithm and the PRESTA-I boundary crossing algorithm were used for all the simulations to enhance the efficacy of the dose calculation (Walters and Kawrakow 2007). The two steps described above are summarized in figure 2.

2.4. Validation of MC simulations

The MC model was validated in terms of the x-ray spectra, the beam quality (i.e. half value layer HVL) measurements, the lateral and axial dose profiles, and the depth dose curves in a previous study (Abuhaimed *et al* 2014). Validation for the PMMA phantoms and $CTDI_{100}$ measurements using a standard 100 mm pencil ionization chamber was also reported in that study. The MC calculations were found to be in good agreement with experimental measurements. As measurements of $CTDI_{100}$ and $f_{100}(150)$ are obtained in a similar manner, the $CTDI_{100}$ validation presented in (Abuhaimed *et al* 2014) was used in this study for $f_{100}(150)$. Further validation was performed for $f(0,150)$ calculations. Dixon and Ballard (2007) assumed that the cumulative dose for a beam of kV energy is measured in terms of air kerma as for $CTDI_{100}$, and this value is used to represent the absorbed dose to the PMMA phantoms, which can be quantified by using a specific protocol such as that introduced by the Institution of Physics and Engineering in Medicine and Biology (IPEMB) code of practice (Aukett *et al* 1996) or the

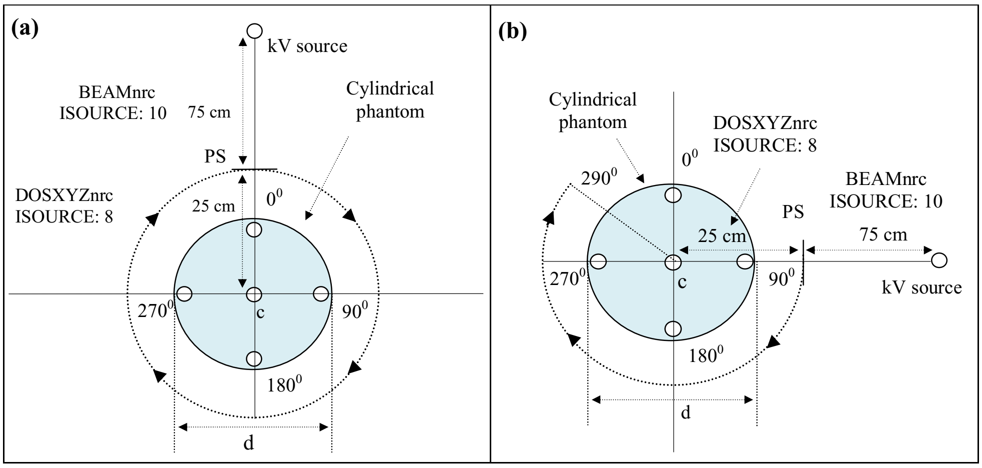

Figure 2. Illustration of the steps used in calculating the absorbed dose within the four phantoms (table 2) using BEAMnrc and DOSXYZnrc. (*a*) representing the FS mode covering 360° (table 1), and (*b*) the HS mode, which covered 200°, from 90° to 290° (table 1).

AAPM protocol (Ma *et al* 2001). Therefore, experimental measurements using the scanning protocols listed in table 1 were performed to measure $f(0,150)$ in terms of air kerma, and the relative results were compared against the absorbed doses obtained from the MC simulations. The measurements for $f(0,150)$ were made with a 0.6 cm^3 Farmer-type ionization chamber (10X5-0.6CT, Radcal Corporation, US) with a calibration traceable to a standard dosimetry laboratory within the standard PMMA phantoms set up at SID = 100 cm. $f(0,150)$ measurements were taken at the middle of the central and four peripheral axes of the phantoms using the clinical beam width W = 198 mm.

2.5. Experimental measurements

$f(0,150)$, $f_{100}(150)$ and $f(0,\infty)$ were measured experimentally using head and body scanning protocols employed in the clinic. The measurements were carried out with the clinical beam width W = 198 mm. Head-HS protocol in table 1 was used for the head scan, and Body-FS was employed for the body scan. $f(0,150)$ measurements were obtained as described in section 2.4, and $f_{100}(150)$ was measured in the same manner as that used for $f(0,150)$, but with using a standard 100 mm pencil ionization chamber (20X6-3CT, Radcal Corporation, US) with a calibration traceable to a standard dosimetry laboratory. $f(0,\infty)$ measurements were obtained within head and body phantoms of length 450 mm. This length is considered sufficiently long to provide a full scatter condition for W = 198 mm (AAPM 2010). Three standard PMMA phantoms were combined together by attaching the ends to form one long phantom. The phantoms were set up at SID = 100 cm and the 0.6 cm^3 Farmer-type ionization chamber (10X5-0.6CT, Radcal Corporation, US) was placed successively at the middle of the central axis and the four peripheral axes to measure $f(0,\infty)$. The experimental measurements of $f(0,150)$, $f_{100}(150)$ and $f(0,\infty)$ were compared with values derived from MC simulation and with different dosimetry approaches CTDI_{100}, the IEC approach ($\mathrm{CTDI}_{\mathrm{IEC}}$) and CTDI_{∞} values which have been investigated in detail in (Abuhaimed *et al* 2014).

Table 3. A comparison between the MC results for $f(0,150)$ and experimental measurements (Exp) obtained within the standard PMMA phantoms using the scanning protocols in table 1 and a beam of width 198 mm. The peripheral positions (p) are shown in figure 2.

	$f(0,150)$—Body-HS		$f(0,150)$—Body-FS		$f(0,150)$—Head-HS		$f(0,150)$—Head-FS	
	Exp	MC	Exp	MC	Exp	MC	Exp	MC
$f(0,150)_c$	**1.00**	**1.00**	0.86	0.85	**1.00**	**1.00**	1.06	1.10
$f(0,150)_{p\text{-}0}$	0.10	0.10	1.20	1.22	0.38	0.34	0.99	1.01
$f(0,150)_{p\text{-}90}$	1.30	1.30	1.20	1.20	0.87	0.84	0.96	1.01
$f(0,150)_{p\text{-}180}$	2.38	2.43	1.19	1.22	1.36	1.41	0.95	0.99
$f(0,150)_{p\text{-}270}$	1.95	1.98	1.21	1.22	1.04	1.07	0.96	1.01
Average p	1.43	1.45	1.20	1.21	0.91	0.90	0.96	1.01
$f(0,150)_w$	1.29	1.30	1.09	1.10	0.94	0.94	0.99	1.03

3. Results

3.1. Validation of MC simulations

Table 3 shows a comparison between $f(0,150)$ values measured experimentally and those obtained from the MC simulations. The values for the body protocols (Body-HS and Body-FS) were normalized with respect to the central value of Body-HS, and values for the head protocols (Head-HS and Head-FS) were normalized with the respect to the central value of Head-HS. The MC values for the body phantom were in good agreement with the experimental measurements, where the average differences for Body-HS and Body-FS were ±0.9% and ±0.8% respectively. However, larger variations were observed for the head protocols, for which the average differences were ±1.1% for Head-HS and ±4.1% for Head-FS. These differences are thought to occur because the Linac couch was not included in the simulations. This omission may have a greater impact on the results for the lower attenuation head phantom, and the differences between the experimental and MC results will be enhanced by the lower tube potential (100 kV) employed for head scans.

3.2. CBCT dosimetry

Values for $f(0,150)_c$, $f_{100}(150)_c$, $f(0,\infty)_c$, and $f_{100}(\infty)_c$ calculated at the centre of the standard and infinitely long phantoms normalized to $\mathrm{CTDI}_{\infty,c}$ values are shown in figure 3. $f(0,150)_c$ and $f_{100}(150)_c$ values within the standard body and head phantoms increased with W until ~150 mm at which point the primary beam began to extend beyond the length of the standard phantoms (150 mm), so the values for $f(0,150)_c$ and $f_{100}(150)_c$ remained virtually constant for $W > 150$ mm (figure 3). The differences between $f(0,150)_c$ and $f_{100}(150)_c$ values for $W = 160$ mm and those obtained for $W = 300$ mm were all within 3%. The long phantoms within which $f(0,\infty)_c$ and $f_{100}(\infty)_c$ calculations were performed, provided a full scatter condition accommodating the whole primary beam and all scattered radiation for $W = 20$–300 mm, and as a result values of $f(0,\infty)_c$ and $f_{100}(\infty)_c$ continued to increase with W.

At a narrow beam width $W = 20$ mm, $f(0,150)_c$ values were larger by 12% and 21% than those for $f_{100}(150)_c$ calculated within the standard body and head phantoms respectively (figure 3). The magnitudes of these differences declined as the beam width increased until $W = 100$ mm, after which the difference stabilized at ~4% for the body phantom and ~5% for the head phantom. The differences between $f(0,\infty)_c$ and $f_{100}(\infty)_c$ were similar to those found

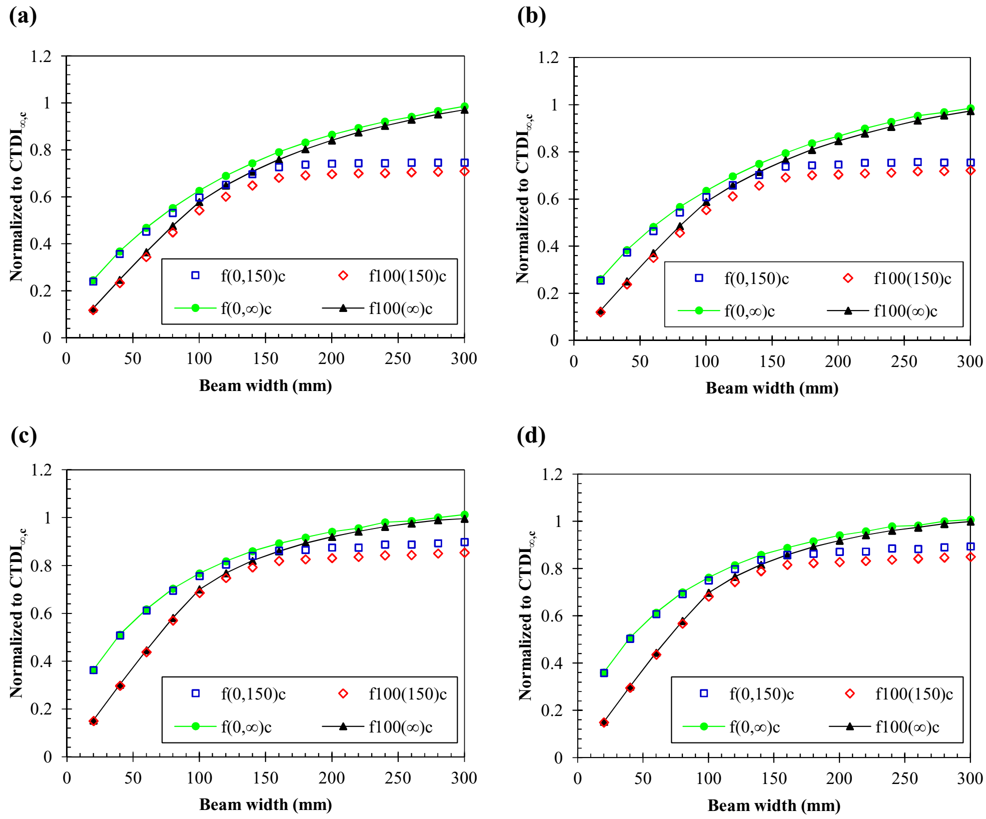

Figure 3. Values for $f(0,150)_c$, $f_{100}(150)_c$, $f(0,\infty)_c$, and $f_{100}(\infty)_c$ normalized to $\mathrm{CTDI}_{\infty,c}$ values calculated at the centre of the body phantoms (*a*) and (*b*), and the head phantoms (*c*) and (*d*) using the scanning protocols (table 1). (*a*) Body-HS, (*b*) Body-FS, (*c*) Head-HS, and (*d*) Head-FS.

between $f(0,150)_c$ and $f_{100}(150)_c$ for $W = 20$ mm, and continued to decline with W reaching 2% and 1% for the body and head phantoms respectively for $W = 300$ mm. In general, the differences between $f(0,150)_c$ and $f_{100}(150)_c$ and between $f(0,\infty)_c$ and $f_{100}(\infty)_c$ in the head phantom were higher than those for the body phantom (figure 3).

Figure 3 shows the difference between contributions resulting from the primary beam and scattered radiation to the cumulative dose measurements under scatter equilibrium and non-equilibrium conditions. For $W \leq 140$ mm, the differences were not as significant, but the differences rose steadily as W increased beyond length of the standard phantom to be 24% between $f(0,150)_c$ and $f(0,\infty)_c$, and 25% between $f_{100}(150)_c$ and $f_{100}(\infty)_c$ for the body phantom (figures 3(*a*) and (*b*)), and 11% and 15% respectively, for the head phantom (figures 3(*c*) and (*d*)) for $W = 300$ mm.

Figure 4 shows $f(0,150)_p$, $f_{100}(150)_p$, $f(0,\infty)_p$, and $f_{100}(\infty)_p$ values calculated at the periphery of the simulated phantoms normalized to $\mathrm{CTDI}_{\infty,p}$ values. $f(0,150)_p$ and $f_{100}(150)_p$ exceeded $\mathrm{CTDI}_{\infty,p}$ values for $W \geq 180$ mm within the standard body phantom, whereas values for the head phantom were lower than $\mathrm{CTDI}_{\infty,p}$ values for all beam widths. In a similar manner to that at the centre of the phantoms, $f(0,150)_p$ and $f_{100}(150)_p$ values remained constant when the beams extended beyond the edges of the phantoms at 3% and 4% for the standard body and head phantoms respectively. As at the centre, $f(0,\infty)_p$ and $f_{100}(\infty)_p$ values increased with W within

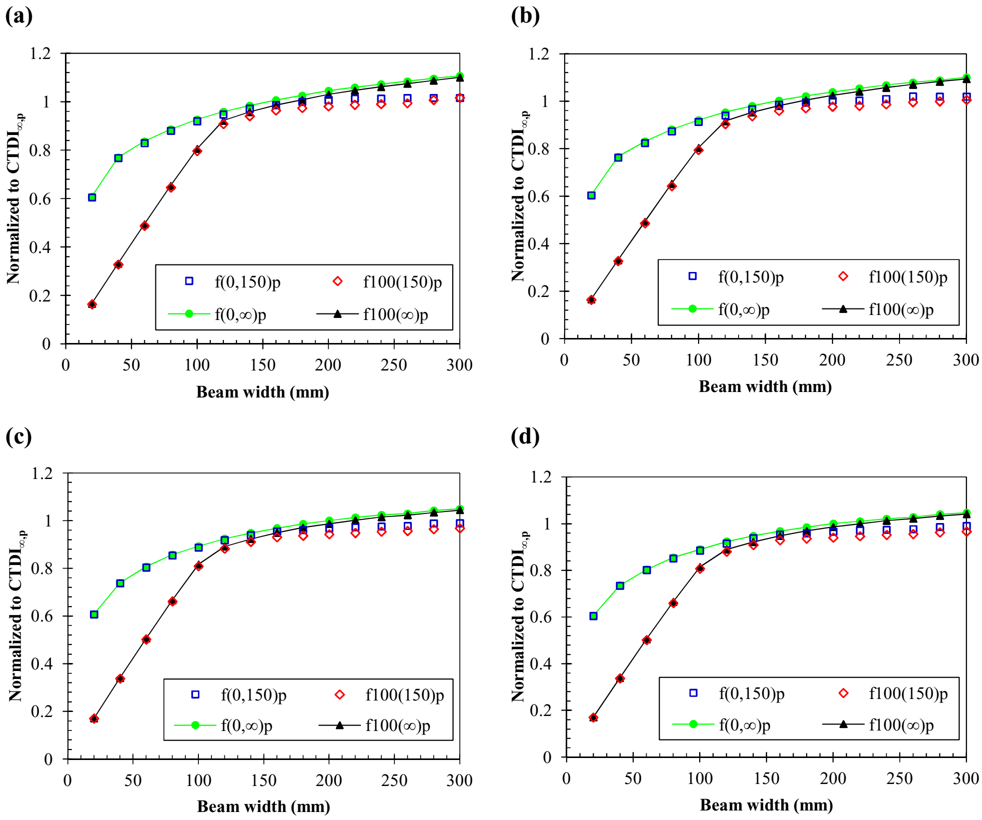

Figure 4. Values for $f(0{,}150)_p$, $f_{100}(150)_p$, $f(0,\infty)_p$, and $f_{100}(\infty)_p$ normalized to $\mathrm{CTDI}_{\infty,p}$ values calculated at the periphery of the body phantoms (*a*) and (*b*), and the head phantoms (*c*) and (*d*) using the scanning protocols (table 1) (*a*) Body-HS, (*b*) Body-FS, (*c*) Head-HS, and (*d*) Head-FS.

all phantoms, but the values exceeded $\mathrm{CTDI}_{\infty,p}$ values for wider beams. $\mathrm{CTDI}_{\infty,p}$ values were overestimated by $f(0,\infty)_p$ and $f_{100}(\infty)_p$ by up to 11% for the body and 5% for the head phantom.

The differences between $f(0{,}150)_p$ and $f_{100}(150)_p$ and between $f(0,\infty)_p$ and $f_{100}(\infty)_p$ for $W \le 100$ mm at the periphery of the phantoms were much larger than those at the centre. For $W = 20$ mm, $f(0{,}150)_p$ was 43% higher than $f_{100}(150)_p$ for the body and head phantoms, and the difference between $f(0,\infty)_p$ and $f_{100}(\infty)_p$ was also 43% for both the phantoms (figure 4). These differences fell with increasing W reaching 2% between $f(0{,}150)_p$ and $f_{100}(150)_p$, and 1% between $f(0,\infty)_p$ and $f_{100}(\infty)_p$ for $W = 300$ mm. The diameter of the phantoms influenced the differences at the centres of the phantoms, but had little effect at the periphery (figures 3 and 4).

The difference between the contributions of the primary beam and scattered radiation to the cumulative dose at the peripheral axes under scatter equilibrium and non-equilibrium conditions was much lower than that at the central axis. For $W = 300$ mm, the differences between $f(0{,}150)_p$ and $f(0,\infty)_p$, and $f_{100}(150)_p$ and $f_{100}(\infty)_p$ were 8% and 10% respectively for the body phantoms (figures 4(*a*) and (*b*)), and 6% and 10% respectively for the head phantoms (figures 4(*c*) and (*d*)).

The weighted values $f(0{,}150)_w$, $f_{100}(150)_w$, $f(0,\infty)_w$, and $f_{100}(\infty)_w$ resulting from the values at the centre (figure 3) and the periphery (figure 4) for the phantoms simulated are shown in

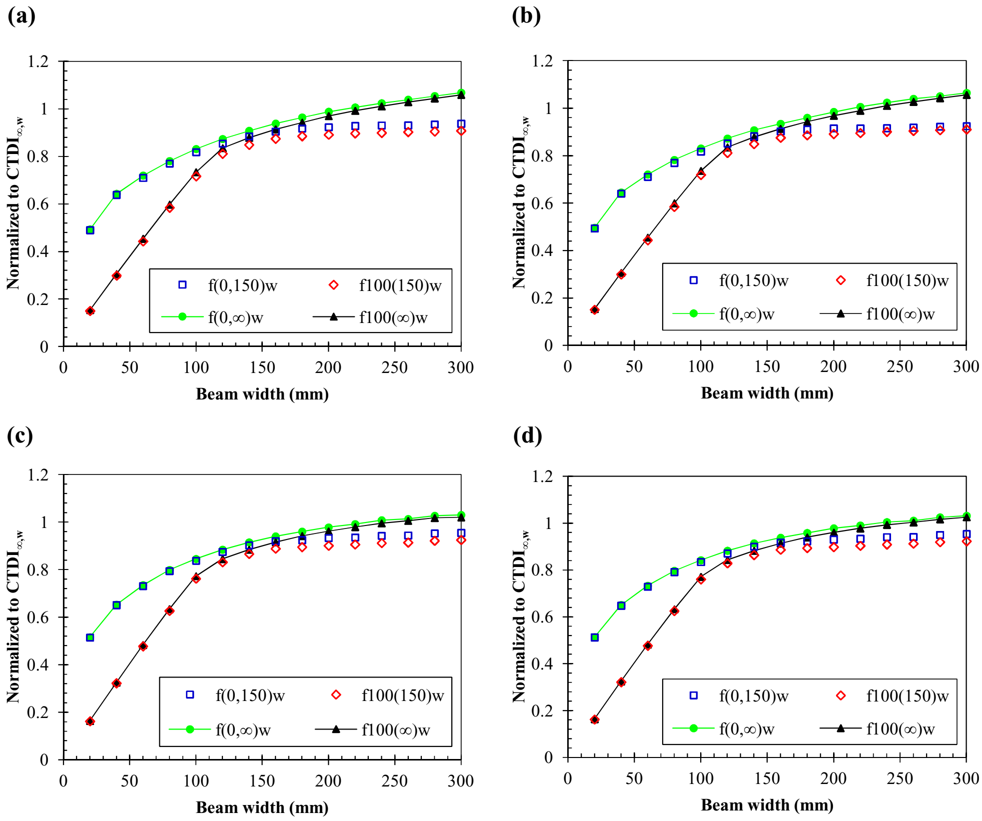

Figure 5. Values for $f(0,150)_w$, $f_{100}(150)_w$, $f(0,\infty)_w$, and $f_{100}(\infty)_w$ normalized to $CTDI_{\infty,w}$ values calculated within the body phantoms (*a*) and (*b*), and the head phantoms (*c*) and (*d*) using the scanning protocols (table 1) (*a*) Body-HS, (*b*) Body-FS, (*c*) Head-HS, and (*d*) Head-FS.

figure 5. From figures 3–5, it can be seen that the scan mode (full or half) has a negligible effect in the dose ratios, where the differences between both the modes at the centre and periphery of the phantoms and the weighted values were all within 1%.

3.3. Experimental measurements

Table 4 shows $CTDI_{100}$, $CTDI_{IEC}$, $f(0,150)$, $f_{100}(150)$ and $f(0,\infty)$ values measured experimentally using the clinical beam width $W = 198$ mm and the head and body scanning protocols. $CTDI_\infty$ values for each scan were estimated by the application of correction factors (C_f) as $CTDI_\infty = C_f \times CTDI_{IEC}$. The correction factors were derived from MC simulations in a previous study (Abuhaimed *et al* 2014). The dose values for all approaches were normalized to the respective $CTDI_\infty$ values at the centre and periphery of the phantoms and the weighted values, and compared to the MC results. The experimental dose ratios for all scanning protocols agreed with the MC dose ratios within 3% (figures 3–5). For all scanning protocols, $f(0,\infty)$ values were higher than those of the other approaches, and provided the closest dose quantities to $CTDI_\infty$ values. $f(0,\infty)_w$ values were within 3% and 2% of $CTDI_{\infty,w}$ values for the head and body scanning protocols respectively.

4. Discussion

4.1. Cumulative dose measurements

In recent years, CBCT dosimetry has received attention from the research community in relation to implementing an appropriate quality assurance (QA) and dosimetry system. The present study investigated the cumulative dose measurements under scatter equilibrium and non-equilibrium conditions. Results for $CTDI_\infty$ were used as standard values for comparison to examine the efficiency of the approaches studied to report $CTDI_\infty$ values. It is recognized that $f(0,150)$ and $CTDI_\infty$ do not represent an exact comparison since one is a measure of point dose and the other an integral of dose from a whole scan. However, the relationship between these variables is an important part of developing a coordinated approach to CBCT dosimetry. The MC results for the approaches studied (figures 3–5) indicated that $f(0,\infty)$ which has been proposed by (AAPM 2010) and is equal to the peak value of a CBCT beam profile provides a good method for estimating $CTDI_\infty$ values. However, the results at the periphery of the phantoms (figure 4) showed that $f(0,\infty)_p$ and $f_{100}(\infty)_p$ overestimated $CTDI_{\infty,p}$ values for $W \geq 160$ mm and $W \geq 180$ mm for the body phantom respectively, and $W \geq 200$ mm and $W \geq 220$ mm for the head phantom respectively. These overestimations resulted in $f(0,\infty)_w$ and $f_{100}(\infty)_w$ values exceeding $CTDI_{\infty,w}$ values within both the phantoms, as the peripheral results have higher weight (2/3) than those at the centre (1/3) (figure 5). This observation is in agreement with findings from a study conducted with $W = 160$ mm using a 320 detector row CT scanner (Aquilion ONE) (Geleijns *et al* 2009). In their study, $f_{100}(150)$ at the periphery of a standard body phantom using 120 kV was a factor of 1.06 greater than $CTDI_{300,p}$ which was taken to represent $CTDI_{\infty,p}$.

The possible reasons for the overestimation may be: (1) the actual beam width which is represented by the full width at half maximum (FWHM) at the periphery of the phantom is lower than the nominal beam width. For example, figure 6 shows FWHM for a beam of a nominal width of 200 mm at the peripheral axes of body and head phantoms. For the body phantom, FWHM = 170 mm at the side from which the x-ray source is incident and FWHM = 230 mm at the other side, and FWHM = 186 mm and 214 mm for the head phantom. This variation in the beam width results from the beam divergence. The diameter of the phantom plays a major role in determining the magnitude of this variation, as this determines the distance of the measuring chamber from the isocentre. The differences between the actual width (FWHM) and the nominal width at the periphery, which is closer to the kV source, for all beam widths $W = 20$–300 mm were ~15% and ~7% for the body and head phantoms, respectively. The main concept for estimating $f(0,\infty)$ and $f_{100}(\infty)$ values differ from that of $CTDI_\infty$, as $f(0,\infty)$ and $f_{100}(\infty)$ are based on averaging the dose over arbitrary chamber lengths regardless of the beam width, whereas $CTDI_\infty$ is based on dividing the dose measured over arbitrary chamber lengths by the nominal beam width. The difference between the FWHM and the nominal width has been shown to have a significant influence on $CTDI_{\infty,p}$ values (figure 4) (Geleijns *et al* 2009). $f(0,\infty)$ and $f_{100}(\infty)$ measurements were also influenced by this difference when the FWHM of the beam at the peripheral axis was less than 20 mm and 100 mm respectively, as the whole chambers were not exposed and the dose was averaged over the lengths. This influence disappeared for wider beams, when the chambers were entirely within the FWHM of the beam. In order to indicate the impact of using the nominal widths and the FWHM on $CTDI_{\infty,p}$ values, figure 7 shows a comparison involving normalization of $f(0,\infty)$ and $f_{100}(\infty)$ values to $CTDI_{\infty,p}$ values calculated using the FWHM instead of the nominal width. The use of the FWHM of the beams studied increased $CTDI_{\infty,p}$ values by 15% and 7% on average for the body and head phantoms respectively, and hence brought the dose ratios below unity (figure 7).

Table 4. Experimental measurements (Exp) for $CTDI_{100}$, $CTDI_{IEC}$, $f_{100}(150)$, $f(0,150)$, and $f(0,\infty)$ for the head and body scanning protocols using the clinical beam width W = 198 mm. The correction factors used to estimate $CTDI_{\infty}$ values were derived from MC simulations and are given in bold and parentheses.

		$CTDI_{100}$[a]	$CTDI_{IEC}$[a]	$f_{100}(150)$	$f(0,150)$	$f(0,\infty)$	$CTDI_{\infty}$
100 kV, 147 mAs				Head scan (mGy)			
Centre		1.72	3.06	3.40	3.51	3.68	4.01
Average periphery		1.62	2.84	3.23	3.21	3.32	3.32
Weighted (1/3 c + 2/3 p)		1.66	2.92	3.29	3.31	3.44	3.56
Ratio to	Exp	0.43	0.76	0.85	0.87	0.91	
$CTDI_{\infty,c}$	MC	0.41	**(1.31)**	0.83	0.89	0.94	
Ratio to	Exp	0.49	0.86	0.97	0.97	1.00	
$CTDI_{\infty,p}$	MC	0.47	**(1.17)**	0.94	0.97	1.00	
Ratio to	Exp	0.47	0.82	0.92	0.93	0.97	
$CTDI_{\infty,w}$	MC	0.45	**(1.22)**	0.90	0.93	0.98	
125 kV, 264 mAs				Body scan (mGy)			
Centre		1.73	2.95	3.40	3.62	4.28	4.81
Average periphery		2.47	4.24	4.86	5.05	5.24	5.1
Weighted (1/3 c + 2/3 p)		2.21	3.81	4.38	4.57	4.92	4.95
Ratio to	Exp	0.36	0.61	0.71	0.75	0.89	
$CTDI_{\infty,c}$	MC	0.35	**(1.63)**	0.70	0.75	0.87	
Ratio to	Exp	0.48	0.83	0.95	0.99	1.03	
$CTDI_{\infty,p}$	MC	0.49	**(1.20)**	0.98	1.00	1.04	
Ratio to	Exp	0.45	0.77	0.88	0.92	0.99	
$CTDI_{\infty,w}$	MC	0.44	**(1.30)**	0.89	0.92	0.98	

[a] Investigated in (Abuhaimed *et al* 2014).

From figure 6, it can be seen that the dose at the centre of the body phantom (figure 6(*c*)) was lower than that at the periphery, and this because of the large diameter of the phantom 32 cm, i.e. more attenuation of the primary beam. However, the dose at the centre of the head phantom (figure 6(*d*)) was higher than that at the periphery, and this is because of the build up of scattered radiation from the wider beam. It has been shown that the dose at the centre of the body phantom was lower than that at the periphery for all scan lengths, whereas the dose at the centre of the head phantom was higher for the scan lengths ≥ ~135 mm (ICRU 2012).

(2) The contribution of scattered radiation to the measurements made at the peripheral axes is lower than to those at the centre. Figures 6(*c*) and (*d*) show the contribution of the scattered radiation resulting from tails of the beam profiles at the centre and periphery of the phantoms. In both the phantoms, the scattered radiation at the centre is higher than that at the periphery, and this increases with increasing the phantom diameter (Boone 2009). The scatter to primary ratio (SPR) values at the periphery of the head and body phantoms were in the range (0.8–1.5) for the x-ray energies (100–130 kV), whereas the SPR values were (3.0–13.0) at the centre of the phantoms for the same x-ray energies (Tsai *et al* 2003, Boone 2009, Li *et al* 2013). The SPR values at the periphery were lower by factors of ~2 and ~7 than those at the centre of the head and body phantoms respectively. When the incident beam becomes wider than the nominal width at the peripheral axis on the side away from the x-ray source (figures 6(*a*) and (*b*)), the beam intensity is lower because of attenuation in the phantom, so that it has less impact on the dose at the periphery. This means that by far the larger component of the dose measurements at the peripheral axes are from the primary beam which has a FWHM lower than the

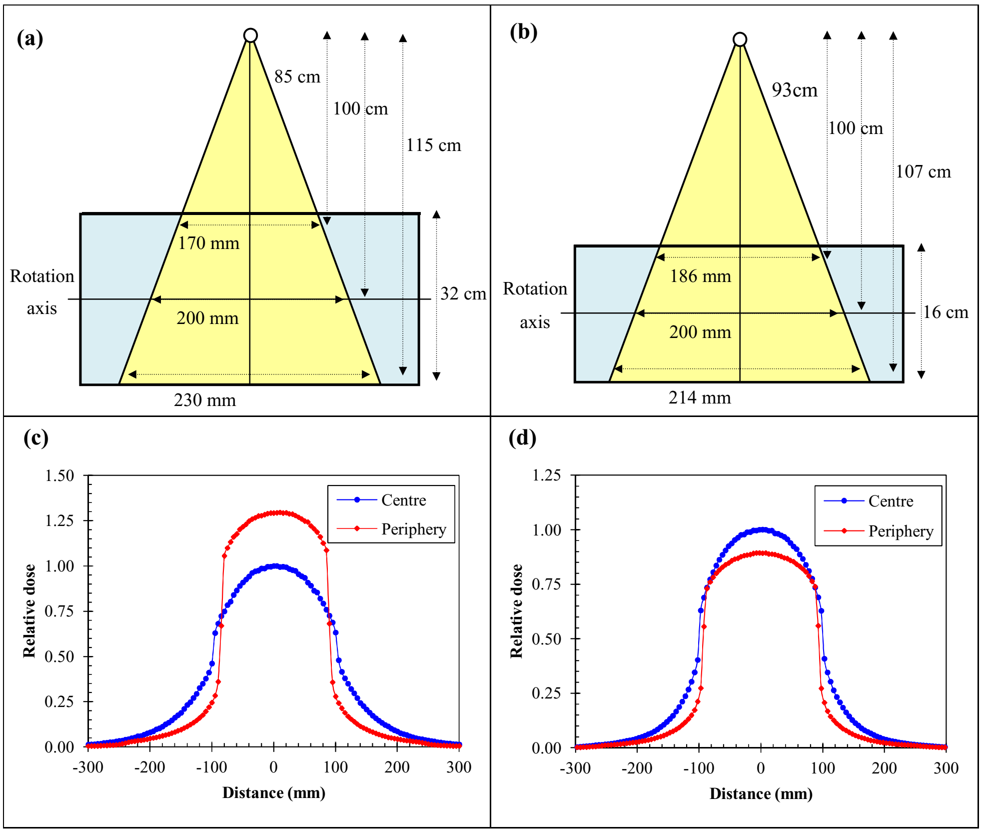

Figure 6. The actual beam width at the peripheral axes of (*a*) body and (*b*) head phantoms for a beam of a nominal width 200 mm. The beam profiles at the centre and periphery of the body (*c*) and head phantoms (*d*) are shown. The dose values in (*c*) and (*d*) are normalized with respect to the central value at (z = 0) of the dose profile of the central axis.

nominal width (figure 6). These two points are thought to be the reasons for the overestimation that occured at the periphery of the phantoms.

4.2. The small and pencil chambers for cumulative dose measurements

The MC results showed that the differences between the small (20 mm) and pencil (100 mm) ionization chambers within the body and head phantoms were large for $W < 120$ mm, but the differences declined when wider beams were used (figures 3–5). At the centre of the phantoms, the differences between $f(0{,}150)_c$ and $f_{100}(150)_c$ as the beam width was increased stabilized after W = 100 mm, and after W = 120 mm at the periphery of the phantoms. This occurred because the 100 mm chamber was not exposed completely at $W < 100$ mm at the centre and < 120 mm at the periphery, whereas the 20 mm chamber was irradiated entirely at $W > 20$ mm. As the $f_{100}(150)$ and $f_{100}(\infty)$ measurements with the 100 mm chamber were averaging the dose over the chamber length, the length of the exposed part played a major role in determining the magnitudes of the measurements, as discussed in section 4.1. However, when the entire 100 mm chamber was irradiated, the differences were reduced, but the 20 mm chamber still gave a higher value (figures 3–5).

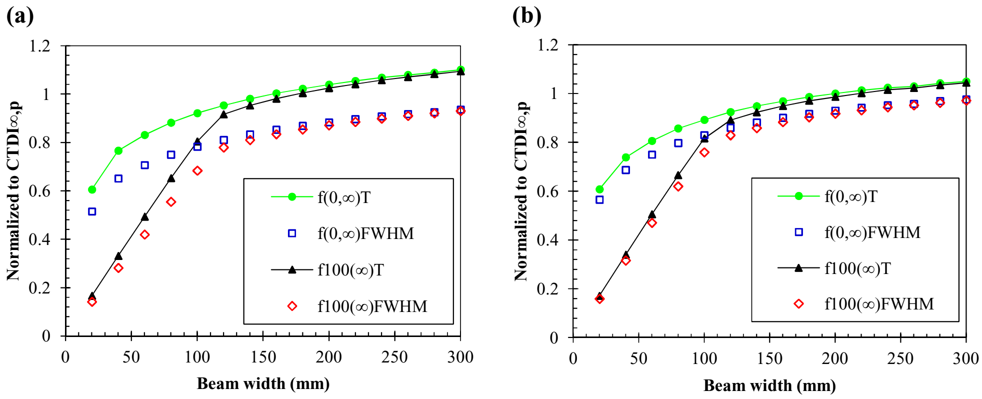

Figure 7. A comparison between the use of the nominal widths (T) and the FWHM of the beams to calculate $\mathrm{CTDI}_{\infty,\mathrm{p}}$ values within the body (a) and head (b) phantoms. T is used for the nominal width instead of W so that it is not confused with the weighted values.

Fahrig *et al* (2006) and Dixon and Boone (2010) have stated that the use of the 100 mm chamber is inappropriate for cumulative dose measurements, which should equate to point dose measurements as proposed by (Dixon 2003). However, results in the present study showed that the differences between measurements obtained with the 20 mm chamber and those of the 100 mm chamber were insignificant for the weighted values for $W \geq 120$ mm within 3% and 4% for the body and head phantoms respectively (figure 5). These findings were in agreement with other studies (table 5), and support the suggestions proposed by (Amer *et al* 2007, Geleijns *et al* 2009, Li *et al* 2014) that the 100 mm chamber be used for the cumulative dose measurements.

4.3. Scatter equilibrium and non-equilibrium conditions

The use of the infinitely long phantoms that provide the scatter equilibrium condition was proposed to measure the cumulative dose (AAPM 2010). The findings in the present study showed that the cumulative dose for $W \leq 140$ mm calculated with the 20 and 100 mm chambers within the infinitely long phantoms did not differ significantly from those calculated within the standard phantoms (figures 3–5), but when wider beams were used, the differences became obvious. As expected, the largest differences between the cumulative dose calculated within the standard phantoms and those within the long phantoms were found at the centre of the body phantoms, as the diameter of the phantom enhances contributions from the scattered radiation to the measurements at the centre as discussed in section 4.1. The influence of the infinitely long phantoms at the peripheral axes was minimal, as the contribution from scattered radiation to the peripheral measurements is much less. The differences between the cumulative dose $f(0,150)$ and $f_{100}(150)$ calculated within the standard body phantom and $f(0,\infty)$ and $f_{100}(\infty)$ calculated within the infinitely long phantoms (figures 3–5) were in good agreement with other studies (table 6).

4.4. Methods for CBCT dosimetry

Since the advent of CBCT, different dosimetric approaches have been proposed. Four of these were investigated in the present study. Although almost all the different international

Table 5. A comparison between dose ratios for the cumulative dose values calculated with the small (20 mm) and pencil (100 mm) chambers in standard and infinitely long head and body phantoms obtained in this study using MC and experimental measurements (Exp) and those published in other studies.

Investigator	Scanner	Method	Tube voltage (kV)	Scanning mode	Phantom	Beam width (mm)	$f(0,150)_c$ / $f_{100}(150)_c$	$f(0,150)_p$ / $f_{100}(150)_p$	$f(0,150)_w$ / $f_{100}(150)_w$	$f(0,\infty)_c$ / $f_{100}(\infty)_c$	$f(0,\infty)_p$ / $f_{100}(\infty)_p$	$f(0,\infty)_w$ / $f_{100}(\infty)_w$
(Li *et al* 2014)	Somatom Definition dual source	MC	100	Full Fan	Head	200	1.05	1.02	1.03	1.02	1.01	1.01
This study		Exp	100			198	**1.03**	**0.99**	**1.01**			
This study		MC	100			200	**1.05**	**1.03**	**1.04**	**1.02**	**1.02**	**1.02**
(Geleijns *et al* 2009)	Toshiba Aquilion ONE	MC	120	Full Fan	Body	160			1.04			1.02
This study		MC	125			160			**1.04**			**1.03**
(Osei *et al* 2009)	Varian On Board Imager	Exp	125	Full Fan	Body	206	1.08	1.03	1.03			
(Li *et al* 2014)	Somatom Definition dual source	MC	120			200	1.07	1.03	1.03	1.03	1.02	1.02
This study		MC	125			200	**1.06**	**1.03**	**1.04**	**1.02**	**1.01**	**1.02**
(Osei *et al* 2009)	Varian OnBoard Imager	Exp	125	Half Fan	Body	206	1.07	1.04	1.05			
This study		Exp	125			198	**1.06**	**1.04**	**1.04**			
This study		MC	125			200	**1.06**	**1.03**	**1.04**			

Table 6. A comparison between dose ratios for the cumulative dose values calculated within standard and infinitely long head and body phantoms obtained in this study using MC and those published in other studies.

Investigator	CT Scanner	Method	Tube voltage (kV)	Phantom	Length of infinitely long phantom (mm)	Beam width (mm)	$f(0,150)_c$ / $f(0,\infty)_c$	$f(0,150)_p$ / $f(0,\infty)_p$	$f(0,150)_w$ / $f(0,\infty)_w$	$f_{100}(150)_c$ / $f_{100}(\infty)_c$	$f_{100}(150)_p$ / $f_{100}(\infty)_p$	$f_{100}(150)_w$ / $f_{100}(\infty)_w$
(Li *et al* 2014)	Somatom Definition dual source	MC	100	Head	900	140–240	0.98–0.91	1.0–0.95	0.99–0.94	0.97–0.88	0.99–0.93	0.98–0.92
This study		MC	100		600	140–240	**0.98–0.90**	**0.99–0.95**	**0.98–0.94**	**0.97–0.87**	**0.99–0.94**	**0.98–0.92**
(Geleijns *et al* 2009)	Toshiba Aquilion ONE	MC	120	Body	700	160			0.98			0.96
This study		MC	125		900	160			**0.97**			**0.96**
(Li *et al* 2014)	Somatom Definition dual source	MC	120	Body	900	140–240	0.92–0.81	0.99–0.95	0.97–0.92	0.90–0.77	0.96–0.94	0.97–0.90
This study		MC	125		900	140–240	**0.94–0.81**	**0.99–0.95**	**0.97–0.91**	**0.92–0.78**	**0.98–0.93**	**0.97–0.89**

Table 7. A comparison between five different approaches for CBCT dosimetry and the standard $CTDI_{100}$. The dose ratios were calculated with beams of width W = 20–300 mm and are shown in figure 8.

		$CTDI_{100}$[a]	$CTDI_{IEC}$[a]	$f(0,150)$	$f_{100}(150)$	$f(0,\infty)$	$f_{100}(\infty)$
Chamber type		100	100	20	100	20	100
Phantom type		Standard	Standard and Free in air	Standard	Standard	Long	Long
Number of measurements		5	7, 8, 9, etc[b]	5	5	5	5
Affected by the nominal beam width (W)		Yes	No [a]	Yes	Yes	Yes	Yes
Availability of equipment worldwide		Yes	Yes	Yes	Yes	No	No
Ratio to	Head	0.75–0.28	0.76 ± 0.01	0.36–0.90	0.15–0.85	0.36–1.01	0.15–1.00
$CTDI_{\infty,c}$	Body	0.59–0.24	0.60 ± 0.01	0.24–0.75	0.12–0.72	0.25–0.99	0.12–0.97
Ratio to	Head	0.85–0.32	0.86 ± 0.01	0.60–0.99	0.17–0.97	0.61–1.05	0.17–1.04
$CTDI_{\infty,p}$	Body	0.82–0.34	0.83 ± 0.01	0.60–1.02	0.16–0.99	0.61–1.11	0.17–1.10
Ratio to	Head	0.81–0.31	0.82 ± 0.01	0.51–0.95	0.16–0.92	0.51–1.03	0.16–1.02
$CTDI_{\infty,w}$	Body	0.75–0.30	0.76 ± 0.01	0.49–0.94	0.15–0.91	0.49–1.07	0.15–1.06

[a] Investigated in (Abuhaimed *et al* 2014).
[b] Number of the measurements required for $CTDI_{IEC}$ is based on the beam width and length of the ionization chamber (IEC 2010, IAEA 2011).

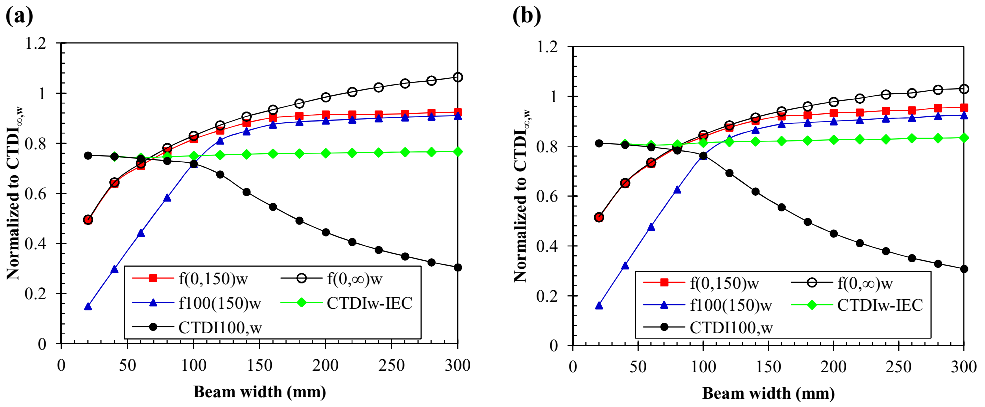

Figure 8. A comparison between dose ratios for four different approaches for CBCT dosimetry and the standard $CTDI_{100}$. The dose values were normalized to $CTDI_{\infty,w}$ values for (*a*) body and (*b*) head phantoms. For clarity, $f_{100}(\infty)_w$ values were not added to the figure, but they are presented in figure 5.

organizations agreed on the failure of $CTDI_{100}$ to be a dose descriptor for CBCT scans, no standard approach has been agreed globally and as a result it is difficult to progress developments in equipment and techniques. Table 7 shows a comparison between the approaches studied as well as the standard $CTDI_{100}$ and $CTDI_{IEC}$ investigated previously in (Abuhaimed *et al* 2014). Each approach has advantages and disadvantages, which make choosing a dosimetric method for CBCT scans difficult. Two main factors that are usually considered when such an issue arises are (1) the efficiency of the approach as a dose descriptor to report the total amount of dose $CTDI_{\infty}$ received by a patient undergoing CBCT scans, and (2) the simplicity of the application of the approach in the clinical environment in terms of availability of the measuring instruments, simplicity of the technique, and the number of the scans required to accomplish a QA assessment, i.e. the QA time. Considering the first factor, figure 8. illustrates the ability of each approach to report the $CTDI_{\infty,w}$ values. All the approaches were strongly dependent on the beam width with the exception of the $CTDI_{IEC}$. However, for $W \geq 100$ mm, the other methods were able to provide values larger than those of $CTDI_{IEC}$, and the higher values were obtained with $f(0,\infty)$ (figure 8). By taking the second factor into account and based on the comparison presented in (table 7), $f(0,150)$ and $f_{100}(150)$ seem to be more practical in the clinic than the other approaches.

Based on the first and second factors, and on the fact that most of CBCT scans are conducted with a beam of width wider than 80 mm, the $f(0,150)$ approach may be the best to be considered for CBCT dosimetry for two reasons: (1) $f(0,150)_w$ values for $W \geq 80$ mm were higher than $CTDI_{IEC,w}$ and $f_{100}(150)_w$ values for the body and head phantoms (figure 5). $f(0,150)_w$ values for $W \geq 80$ ranged from 0.77 to 0.92 of $CTDI_{\infty,w}$ values within the body phantom, and from 0.79 to 0.95 within the head phantom. $f(0,150)_w$ values were approximately stable for $W > 150$ at ~0.92 and ~0.94 for the body and head phantoms respectively. These underestimations, however, can be overcome by the application of correction factors. (2) Although $f(0,\infty)_w$ and $f_{100}(\infty)_w$ values were higher than those of $f(0,150)_w$, the $f(0,150)$ approach is more practical in the clinical environment in terms of the handling and availability of equipment and the QA measurement time.

The results in this study and those presented in (Abuhaimed *et al* 2014) have been shown to be in good agreement with other investigations that were conducted with different CT and CBCT scanners. This agreement gives an indication that the dose ratios shown in all studies may be independent of the kV device. Moreover, it has been shown that the dose ratios have relatively weak dependencies on tube voltages (80–140 kV) and the bowtie filter type (head and body) (Li *et al* 2014). Thus, the dose ratios of this study may be comparable for other CT and CBCT scanners.

5. Conclusion

Four approaches $f(0,150)$, $f_{100}(150)$, $f(0,\infty)$ and $f_{100}(\infty)$ have been investigated in the present study using MC EGSnrc-based user codes BEAMnrc and DOSXYZnrc. The kV system integrated with a TrueBeam linear accelerator was simulated and the results were benchmarked against experimental measurements in a previous study as well as this study. The dose values for the approaches studied were calculated using four different scanning protocols, and the results were compared to CTDI_∞ values. The $f(0,\infty)$ approach gave the highest dose values compared with the other approaches, and its values were the closet to CTDI_∞ values especially for wider beams. Dose values obtained with the 20 and 100 mm chambers for the cumulative dose measurements were substantially different for $W < 120$ mm, but the differences fell with increasing beam width and became approximately comparable. The need for infinitely long phantoms to be used for the cumulative dose measurements was not significant for $W \leq 140$ mm, but such phantoms are required for wider beams to account for all the contributions arising from the primary beam and the scattered radiation that cannot be captured within the standard phantoms. By comparing five different approaches for CBCT dosimetry in terms of the efficiency of the approach as a dose descriptor and the simplicity of the implementation in the clinical environment, the $f(0,150)$ approach may be the best for CBCT dosimetry, using correction factors to adjust the values reported to give indicators closely linked to patient dose.

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